

SUJAY KUMAR MONDAL

Python Backend Developer | FastAPI | REST APIs | PostgreSQL | Redis

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PROFESSIONAL SUMMARY

Experienced Python Backend Developer with 2+ years of hands-on expertise building scalable RESTful APIs and production-grade backend systems using FastAPI. Adept at designing clean API architectures, integrating third-party services, and implementing efficient database models with PostgreSQL and MySQL. Strong foundation in API security, JWT/OAuth2 authentication, and event-driven backend design using Redis and Kafka. Proven ability to collaborate in fast-paced environments and deliver high-quality, production-ready code.

TECHNICAL SKILLS

Languages	Python
Frameworks	FastAPI, Django, SQLAlchemy, Pydantic
Databases	PostgreSQL, MySQL — schema design, query optimisation, indexing
Caching / MQ	Redis (Streams, pub/sub, caching), Apache Kafka, Redis Streams
Auth & APIs	JWT, OAuth2, OAuth SSO (Google, GitHub), RESTful API design
Architecture	Microservices, gRPC, event-driven design, async / concurrent patterns
DevOps	Git, CI/CD, Docker, Uvicorn, Gunicorn

PROFESSIONAL EXPERIENCE

System Engineer — Python Backend Developer | Surelia Infosystems Private Limited Aug 2023 – Present

- Architected scalable RESTful APIs with FastAPI for multiple production applications serving concurrent user loads.
- Designed and optimised PostgreSQL and MySQL schemas using SQLAlchemy ORM; strategic indexing and query refactoring reduced average query latency by 25%.
- Built modular microservices with FastAPI and gRPC, improving service isolation, scalability, and independent deployability across the platform.
- Integrated Apache Kafka and Redis Streams for asynchronous, event-driven inter-service communication in high-throughput data pipelines.
- Implemented JWT/OAuth2 authentication with RBAC middleware and third-party SSO (Google, GitHub) across all production services.
- Defined API contracts and versioning standards with client teams, reducing deployment-related incidents in cross-team integrations.

KEY PROJECTS

AI Chatbot — Backend API Service — FastAPI, PostgreSQL, SQLAlchemy, JWT, Redis, Python

- Designed the full backend: async FastAPI routes, PostgreSQL schema for session-scoped multi-user chat history, and SQLAlchemy ORM models with indexed queries for efficient conversation retrieval.
- Built JWT-based stateless auth with token refresh and user-scoped access control; integrated external AI model APIs as async background tasks to decouple inference latency from HTTP responses.
- Structured modular FastAPI routers (auth, chat, sessions) with Pydantic schema validation and consistent error handling, following RESTful API design best practices.

Authentication & Authorisation Platform — FastAPI, OAuth2, JWT, MySQL, SQLAlchemy, bcrypt

- Developed standalone auth microservice with OAuth2 password flow, JWT access/refresh token lifecycle, bcrypt hashing, and RBAC middleware; designed MySQL schema for users, roles, and sessions.
- Integrated Google and GitHub OAuth SSO, abstracting provider-specific callback flows behind a unified FastAPI endpoint.

E-Commerce & Market Data Platform — FastAPI, Redis Streams, MySQL, Kafka, gRPC, Docker

- Built a Docker-deployed microservice ecosystem; used Redis Streams as an event bus (thousands of events/min with consumer group acknowledgement) and Kafka for durable message queuing.
- Implemented gRPC for synchronous internal service calls; optimised MySQL schemas with composite indexes for high-load catalogue and real-time market tick data queries.

EDUCATION

MCA (Master of Computer Applications)

2021 – 2023

Brainware University • 88.74%

B.Sc. in Computer Science

2018 – 2021

Calcutta University • 65%